

BELBIN BENO R M

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PROFESSIONAL PROFILE

AI Engineer with a **B.E. in Computer Science (8.53 CGPA)** and a professional background in **Quantitative Analysis** and **Strategic Resource Planning**. Specialized in building **RAG-based Generative AI pipelines**, high-precision **Stacking Ensembles (0.9536 AUC)**, and automated ML utilities. Expert in Python (OOP), FastAPI, and SQL, with a proven ability to modernize industrial workflows into scalable AI solutions. Available for immediate joining.

CORE TECHNICAL SKILLS

- **Generative AI & NLP:** RAG (Retrieval-Augmented Generation), LangChain, Vector Databases (pgvector), LLM API Integration (OpenAI, Gemini, Groq), Prompt Engineering.
- **Machine Learning:** Supervised Learning (XGBoost, LightGBM, CatBoost), Stacking/Voting Ensembles, Hyperparameter Tuning (Optuna), Scikit-Learn, TensorFlow.
- **Engineering & Deployment:** FastAPI, LLM, RESTful API Development, Docker (Basics).
- **Data Engineering:** Python (Expert/OOP), SQL (PostgreSQL), ETL Pipelines, Feature Engineering, Automated Data Validation.
- **Tools:** Git/GitHub, Power BI, n8n Automation, Advanced Excel.

PROFESSIONAL EXPERIENCE

Quantitative Analyst | Adroit Design India Pvt Ltd – Chennai

May 2024 – Feb 2026 (1 Year 10 Months)

- **Strategic Resource Planning:** Managed end-to-end project analytics and strategic planning for large-scale infrastructure and construction projects.
- **Cost Estimation & Analytics:** Developed data-driven models for cost estimation, ensuring high-integrity reporting for project budgeting.
- **Workflow Automation:** Transitioned legacy Excel-based civil engineering workflows into automated Python cleaning and analytical scripts, reducing manual processing time by 40%.
- **KPI Reporting:** Audited high-volume data streams to deliver monthly KPI reports that optimized resource allocation and material usage trends.
- **Technical Integration:** Bridged the gap between raw construction data and downstream BI dashboards for executive decision-making.

Machine Learning Practitioner | Kaggle (Competitive Modeling)

2023 – Present

- **Ensemble Optimization:** Consistently applied Stratified K-Fold cross-validation and nested stacking techniques to structured datasets to achieve top-tier AUC scores.
- **Model Explainability:** Developed reproducible Jupyter Notebooks focusing on feature selection, importance gain, and model interpretability (SHAP/LIME).
- **Feature Engineering:** Specialized in handling high-kurtosis data and extreme outliers using advanced clipping and transformation strategies.

KEY PROJECTS

AegisAI: Production Agent Platform & Governance

- Built a production AI platform with a central Main Agent orchestrating reasoning, multi-turn memory, and Model Context Protocol (MCP) tools.
- Implemented a pgvector-backed hybrid search RAG pipeline and an asynchronous document ingestion system using Upstash Redis and background workers.
- Developed controlled MCP tool interfaces for semantic retrieval, chat history, and live web search.
- Enforced application-level guardrails, including execution step limits, tool call bounds, and automated clarification workflows.

RAGForge: Knowledge Retrieval & Agent Pipeline

- Developed an AI knowledge platform supporting multi-format document ingestion (PDF, DOCX, TXT) with custom text chunking.
- Implemented semantic vector retrieval using Sentence Transformers and PostgreSQL/pgvector with configurable similarity thresholds.
- Integrated a Gemini-powered agent featuring dynamic MCP tool discovery and execution for grounded document responses.
- Built persistent session-aware conversational memory backed by PostgreSQL REST endpoints and FastAPI.

QLoRA Fine-Tuning: Cardiovascular Risk Adaptation

- Fine-tuned an open-weights causal language model using 4-bit NormalFloat (NF4) quantization via BitsAndBytes for parameter-efficient adaptation.
- Applied Low-Rank Adaptation (LoRA) target modules to adapt the model to structured clinical patient records.
- Engineered instruction-formatted prompt pipelines to transform tabular patient attributes into Supervised Fine-Tuning (SFT) targets.
- Executed SFT training using gradient accumulation and mixed-precision arithmetic, tracking perplexity and ROC-AUC metrics.

AI Customer Support Platform & Agent Workflows

- Built a FastAPI support agent platform delivering grounded Q&A over knowledge-base content using RAG.
- Implemented multi-turn conversation memory and function-calling tools for real-time customer profile and order lookups.
- Developed an automated ticket management API featuring escalation workflows and action audit logs. Containerized the platform with Docker Compose and integrated an automated Pytest test suite.

EDUCATION

- **B.E. Computer Science Engineering** | Anna University | 2024 | **CGPA: 8.53**
- **PGP – Data Analytics & Data Science** | Skill-Lync | 2024